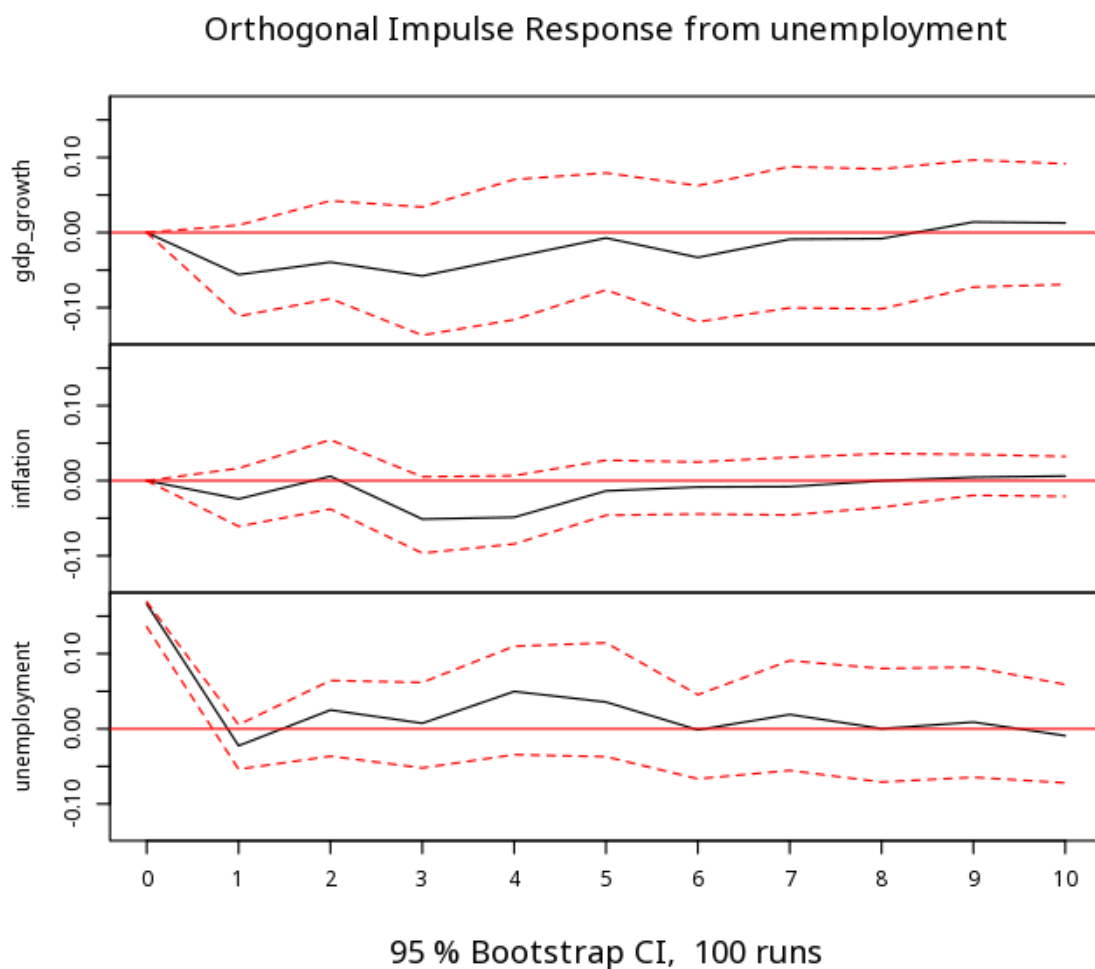


1 VAR

Lag	AIC	BIC	HQ
1	-8.25	-7.96	-8.13
2	-8.25	-7.73	-8.04
3	-8.23	-7.50	-7.93
4	-8.20	-7.24	-7.81
5	-8.30	-7.12	-7.82
6	-8.22	-6.82	-7.65
7	-8.21	-6.59	-7.55
8	-8.14	-6.30	-7.39
9	-8.10	-6.04	-7.27
10	-8.13	-5.85	-7.21

	gdp_growth	inflation	unemployment
gdp_growth.l1	0.44 (0.11) [0.22, 0.67]	0.22 (0.07) [0.07, 0.37]	-0.82 (0.05) [-0.91, -0.72]
inflation.l1	0.18 (0.18) [-0.18, 0.54]	0.09 (0.12) [-0.15, 0.32]	0.16 (0.08) [0.00, 0.32]
unemployment.l1	-0.34 (0.23) [-0.80, 0.13]	-0.15 (0.15) [-0.45, 0.15]	-0.14 (0.10) [-0.34, 0.06]
gdp_growth.l2	0.26 (0.23) [-0.19, 0.71]	-0.05 (0.15) [-0.35, 0.24]	-0.11 (0.10) [-0.31, 0.08]
inflation.l2	0.00 (0.19) [-0.37, 0.37]	0.19 (0.12) [-0.05, 0.43]	-0.02 (0.08) [-0.18, 0.14]
unemployment.l2	-0.11 (0.23) [-0.57, 0.36]	0.10 (0.15) [-0.20, 0.40]	-0.12 (0.10) [-0.32, 0.08]
gdp_growth.l3	0.11 (0.23) [-0.35, 0.56]	0.07 (0.15) [-0.23, 0.37]	-0.12 (0.10) [-0.31, 0.08]
inflation.l3	-0.12 (0.18) [-0.48, 0.24]	0.12 (0.12) [-0.11, 0.36]	0.12 (0.08) [-0.03, 0.28]
unemployment.l3	-0.12 (0.23) [-0.59, 0.34]	-0.21 (0.15) [-0.51, 0.09]	-0.19 (0.10) [-0.39, 0.01]
gdp_growth.l4	-0.00 (0.23) [-0.45, 0.45]	-0.13 (0.15) [-0.43, 0.16]	-0.14 (0.10) [-0.34, 0.06]
inflation.l4	-0.06 (0.18) [-0.43, 0.30]	-0.05 (0.12) [-0.29, 0.19]	0.02 (0.08) [-0.14, 0.17]
unemployment.l4	0.11 (0.24) [-0.36, 0.58]	-0.21 (0.15) [-0.51, 0.10]	0.02 (0.10) [-0.19, 0.22]
gdp_growth.l5	-0.09 (0.22) [-0.53, 0.35]	-0.22 (0.14) [-0.50, 0.07]	0.03 (0.10) [-0.16, 0.22]
inflation.l5	0.01 (0.18) [-0.35, 0.37]	-0.11 (0.12) [-0.35, 0.12]	0.06 (0.08) [-0.10, 0.22]
unemployment.l5	0.35 (0.11) [0.13, 0.56]	0.03 (0.07) [-0.11, 0.17]	0.06 (0.05) [-0.04, 0.15]
const	1.14 (3.25) [-5.30, 7.58]	4.30 2(2.11) [0.11, 8.49]	7.61 (1.40) [4.83, 10.39]
Num.Obs.	115	115	115
R ²	.88	.70	.96
R ² Adj.	.87	.65	.96
RMSE	0.36	0.24	0.16
Log.Lik.	-46.53	2.89	50.06
Selected lag (p) = 5			

Variable	Impulse	Share
gdp_growth	gdp_growth	.99
gdp_growth	inflation	.00
gdp_growth	unemployment	.01
inflation	gdp_growth	.68
inflation	inflation	.29
inflation	unemployment	.04
unemployment	gdp_growth	.93
unemployment	inflation	.01
unemployment	unemployment	.06



A VAR models several series as functions of their joint past. The impulse-

response plots show how a shock to one series propagates to the others over time; read each IRF together with its bootstrap confidence band, and note that orthogonalised (Cholesky) IRFs depend on the ordering of the series. Lag selection picks how many past periods to include. A level VAR assumes stationary series — difference them (or use a VECM) if they are not. The Portmanteau test (`vars::serial.test`) checks the residuals for remaining serial correlation: a small p suggests the lag order is too low to whiten the residuals. Method: R vars package (Pfaff, 2008; Lütkepohl, New Introduction to Multiple Time Series Analysis).

A VAR(5) model was selected by AIC; impulse-response functions are shown (Figure).

Statistical basis: Vector autoregression (VAR) estimation (Pfaff B (2008). "VAR, SVAR and SVEC Models: Implementation Within R Package vars." *Journal of Statistical Software*, 27(4).); VAR modeling framework and impulse-response interpretation (Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer-Verlag, Berlin.).